CivicFix AI

Geo-Aware Urban Infrastructure Complaint Prioritization

Assignment 2 / Updated Project Report • Group 8 • 29 October 2026

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Executive summary

CivicFix AI prioritizes municipal road, sidewalk and drainage complaints using hazard severity, recurrence risk and rainfall. In response to instructor feedback about transferring a model trained on New York City (NYC) data to Istanbul, this update adds an Istanbul-specific rainfall case study and a Copernicus DEM terrain-screening pilot. The NYC recurrence model has been rerun on the supplied 2025 data. The Istanbul component is an exploratory geographic prototype, not a validated Istanbul flood-prediction model.

1. Problem and research questions

The project supports SDG 11.5 by helping maintenance planners identify urgent infrastructure complaints. RQ1: Can complaint history predict recurrence within 30 days? RQ2: Can short informal Turkish complaints be categorized by issue and hazard tier? RQ3: Can geography and local rainfall improve Istanbul-specific prioritization? RQ4: Does the resulting queue prioritize critical complaints more effectively than arrival order? Only RQ1 and the exploratory geographic preparation for RQ3 are empirically addressed in this delivery.

2. Datasets and processing

Dataset	Coverage	Use
NYC 311 complaints	2025; 132,648 raw / 132,506 cleaned	Training and chronological evaluation
NOAA Central Park	Daily 2025 precipitation	NYC rainfall features
Open-Meteo Istanbul	1–20 Oct 2026; 41.0193°N, 28.8873°E	Single-point event rainfall
Copernicus DEM COP30	28.9118–29.0424°E; 41.0310–41.1010°N	Pilot terrain analysis

The Istanbul precipitation coordinate is outside the downloaded DEM footprint. They therefore cannot be interpreted as spatially coincident observations. The older IBB reservoir rainfall dataset was reviewed but is not used for the October 2026 event analysis.

3. NYC recurrence model — reproduced evaluation

Target: another complaint in the same approximately 0.001-degree grid cell within 30 days. Training uses complaints before 1 September 2025; testing covers September–November 2025. December is excluded to avoid unobserved 30-day outcomes. Inputs include complaint type, descriptor, hazard tier, borough, location type, calendar variables, current and preceding 3-day rainfall, and past complaint counts. The implementation uses HistGradientBoostingClassifier (120 iterations); these settings differ from the original assignment, so scores are not a bit-for-bit replication.

Metric	Result
Training observations	76,437
Test observations	27,937
ROC-AUC	0.672
PR-AUC	0.585
F1 at threshold 0.50	0.474
Top-decile precision	71.6%
Test recurrence prevalence	39.9%

Top-decile precision means that about 71.6% of the highest-scored 10% of evaluated NYC complaints were followed by another complaint within 30 days, versus 39.9% in the overall test set. It does not measure flood occurrence or Istanbul performance.

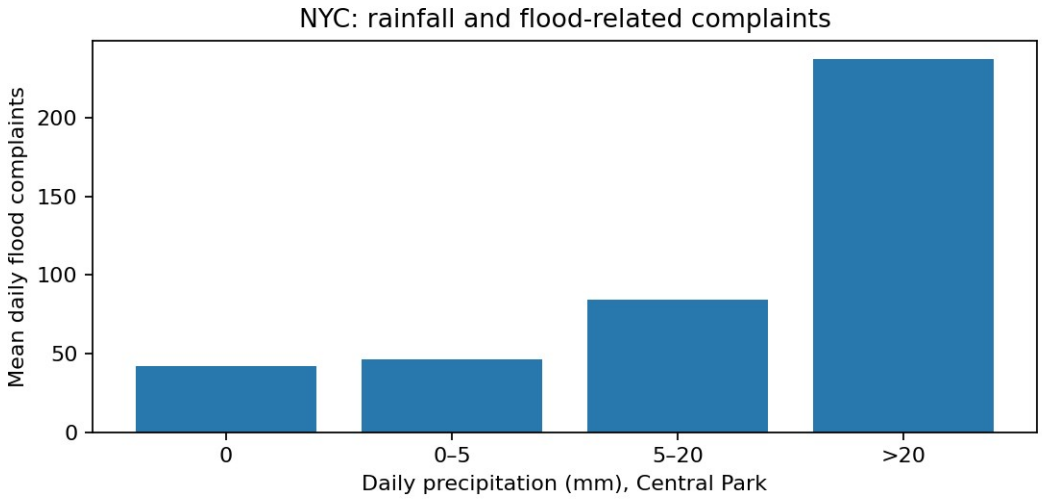


Figure 1. NYC rainfall bins and mean daily flood-related complaint counts (descriptive association, not causality).

4. Istanbul rainfall case study

The supplied Open-Meteo series for 1–20 October 2026 shows 167.8 mm of daily precipitation on 14 October at the selected Istanbul coordinate. This is a location-specific modeled/reanalysis weather series; it is not an Istanbul-wide measured total. No claim is made that this rainfall directly caused any particular complaint or flood extent.

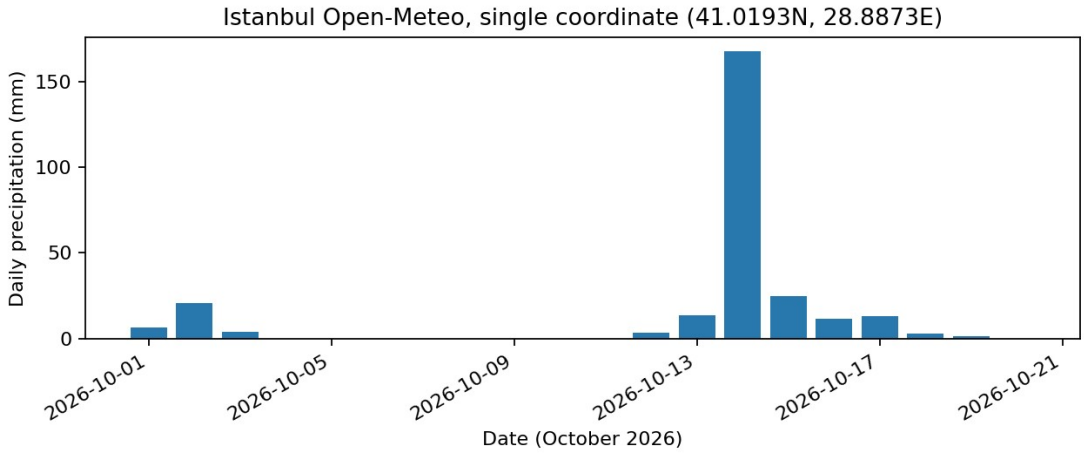


Figure 2. Daily precipitation at one Istanbul location, 1–20 October 2026.

5. Geographic terrain-screening prototype

The Copernicus DEM was processed to derive terrain slope and a local-relative-elevation proxy. A transparent exploratory index (0–100) combines 55% low local relative elevation and 45% low slope, each clipped to a 0–1 range. Local relative elevation is computed against a 15-pixel neighborhood minimum. Higher index values identify terrain warranting further investigation; they do NOT represent a calibrated probability of flooding. The method does not include drainage networks, impervious surface, hydrological flow accumulation, culverts or underground/covered streams.

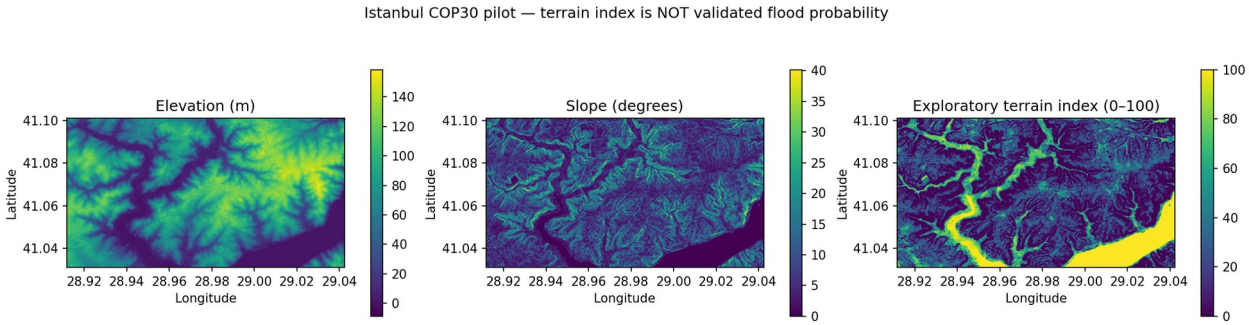


Figure 3. COP30 pilot area: elevation, slope and exploratory terrain index. Coverage is only the downloaded geographic window, not all of Istanbul.

6. Instructor feedback: response and outstanding work

Istanbul-specific data: Added Open-Meteo rainfall and COP30 elevation data. No Istanbul record-level complaint labels were supplied, so transfer performance cannot yet be tested.

Geographic novelty: Implemented a reproducible terrain index using DEM. Adding OpenStreetMap stream features, verified buried stream channels and flow accumulation is future work.

Turkish-language LLM: A 200-item anonymized, hand-labeled Turkish complaint benchmark remains planned. No LLM classification experiment has been executed or claimed here.

Satellite validation: Sentinel-1 flood extent extraction was not performed. Urban radar water detection and timing would require separate validation.

Cross-city bridge: Terrain features can be computed with identical definitions for NYC and Istanbul; this has not yet been implemented for NYC or evaluated for domain transfer.

Equity and privacy: District-level underreporting and ranking fairness remain unevaluated. Real citizen complaint text must be de-identified before use in external APIs.

7. Conclusions and next steps

The supplied NYC model remains a promising recurrence-ranking baseline (ROC-AUC 0.672; top-decile precision 71.6%). The Istanbul-specific contribution is now a reproducible weather and terrain analysis that makes the intended localization pathway concrete. However, no valid NYC-versus-Istanbul predictive improvement can be claimed without geolocated Istanbul complaint/flood labels. The highest-priority next step is to obtain such records, spatially join them with terrain and precipitation, and evaluate against a time-separated Istanbul holdout.

References and provenance

- NYC Open Data, 311 Service Requests (dataset erm2-nwe9), 2025 export; https://data.cityofnewyork.us/resource/erm2-nwe9
- NOAA NCEI GHCN-Daily, Central Park station USW00094728, supplied 2025 CSV.
- Open-Meteo Historical Weather, user-supplied October 2026 workbook; https://open-meteo.com/en/docs/historical-weather-api
- Copernicus DEM GLO-30, user-supplied OpenTopography raster archive; https://portal.opentopography.org/
- Original CivicFix AI Assignment 1 report and civicfix_eda.py, Group 8 (supplied project materials).

Reproducibility: run_analysis.py, results.json and exported PNG figures accompany this report. Figures and values were computed from the supplied local datasets.